

Conart-MSM Sachet for Oral Solution

Name and Strength of Active Substances

Crystalline Glucosamine sulphate sodium chloride 1884mg equivalent 1500mg glucosamine sulphate (derived from Seafood), Chondroitin sulphate 1200mg (bovine origin) and Methylsulfonylmethane 1500mg

Product Description

White to off-white powder to be dissolved in water and form a clear solution with lemon flavour.

Pharmacodynamics

Glucosamine is a natural substance found in chitin, mucoproteins, and mucopolysaccharides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body; glucosamine is also present in tendons and ligaments. Glucosamine must be synthesised by the body but the ability to do this declines with age. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis. Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity. The primary mechanism of action of CS is increased joint glycosaminoglycan concentration and a subsequent enhancement of synovial fluid viscosity. Improvement in joint structure and function are due to: (1) increased endogenous synthesis of hyaluronic acid and sulfated glycosaminoglycans from chondroitin sulfates; and (2) reduced breakdown of joint glycosaminoglycans subsequent to decreased collagenolytic activity and inhibition of enzymes, such as phospholipase A2 and N-acetylglucosaminidase, which are capable of degrading existing joint glycosaminoglycans. Methylsulfonylmethane (MSM) is an organic sulphur – containing compound that occurs naturally in a variety of fruits, vegetables, grains and animals including human. It blocks the inflammatory process and enhances the activity of cortisol, a natural anti-inflammatory hormone produced in the body. MSM is an effective natural analgesic and has been used as an application for pain relief over arthritis joints. Glucosamine and chondroitin work synergistically to stimulate the anabolic metabolism of osteo-cartilaginous tissues by stimulating the biosynthesis of the mucopolysaccharides (which are the essential components of the cartilage ground substance). Addition of methylsulfonylmethane strengthen the cartilage structure and help to reduce the inflammatory process associated with the arthritic joints.

Pharmacokinetics

Absorption

After oral administration, bioavailability is low due to first-pass hepatic metabolism~26%
The gastrointestinal absorption is close to 90%.

In a study of oral chondroitin sulphate in healthy male volunteers, plasma levels of chondroitin sulphate increased with a peak concentration after 2 h. The bioavailability ranges from 20% to 85%. From the absorbed fraction only 10-15% is in Chondroitin sulphate form, that is exists an important first pass effect. Absorption seems to be proximal sections rather than distal sections of the digestive tract. Metabolism and excretion of MSM found the primary metabolism became detectable in serum approximately 2 hours

after ingestion of MSM. With continued MSM ingestion, metabolite maintained a steady concentration in the serum. When MSM was stopped after 14 days, the mean metabolite concentration declined slowly over the subsequent 96 hours, and only trace amount was detectable after 5 days. The decline serum metabolite was linear, and its half-life appeared to be about 38 hours. Metabolite has been shown to persist in the blood up to 5 times longer than MSM.

Distribution

Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins). Volume of Distribution: 2.5 Liters.

Chondroitin is plasma protein-bound.

Metabolism

Glucosamine: Liver, extensive

The first-pass effect in the liver in which more than 70% of glucosamine is metabolized.

Chondroitin: Liver

MSM: Gut or liver

Excretion

Glucosamine: Renal Excretion, 10%; feces, 11%

Part of a dose of glucosamine sulphate is eliminated as carbon dioxide via expired air

Chondroitin: mainly through urine and feces. The $t_{1/2}$ elimination is about 10hours

MSM: Mainly via urine.

Indication

Adjuvant therapy for osteoarthritis.

Recommended Dosage

Adults

One sachet to be taken daily before meal. The entire contents of one sachet should be fully dissolved in at least 250 ml of water (one glass) and drink immediately. It is best to be taken before meal for at least 8-12 weeks or more, according to medical prescription. The treatment of osteoarthritis should be repeated every other 6 months or less (according to medical prescription).

Children

Safety and effectiveness have not been established in children

Mode of Administration

Oral

Contraindications

Hypersensitivity to glucosamine or to chondroitin or methylsulfonylmethane or to any of the excipients
As the active ingredient glucosamine is derived from seafood, the product should not be given to patients who are allergic to seafood

Warnings and Precautions

- Conart-MSM treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2-3 weeks. Therefore, it is advisable to take an analgesic/anti-inflammatory drug if required during the first 2-3 weeks of therapy with glucosamine.
- Chitosan may form complexes with chondroitin sulphate, decreasing absorption. Therefore do not use concomitantly with

chitosan.

- Administration during the first three months of pregnancy must be avoided.
- Safety and effectiveness have not been established in children therefore children, should avoid using glucosamine and chondroitin.
- The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision.
- Since one of its ingredient ,i.e. glucosamine sulphate is derived from seafood, Conart-MSM sachet should not be given to patients who are allergic to shellfish.

A doctor should be consulted in order to exclude the presence of other joint conditions/diseases for which an alternative treatment should be considered

Effects on Ability to Drive and Use Machines

No effects on the ability to drive or to operate machines are expected.

Interactions with Other Medicaments

Effects on glucose metabolism & antidiabetic agents:

It has been hypothesized that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of peripheral glucose uptake. Glucosamine may increase insulin resistance and consequently affect glucose tolerance. It may reduce antidiabetic agent effectiveness when used with these antidiabetic agents: Acarbose, Acetohexamide, Chlorpropamide, Glipizide, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride, Tolbutamide, Troglitazone, Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely/more frequently

Reduced effectiveness when used with glucosamine: Doxorubicin , Etoposide , Teniposide

Warfarin

- elevations of International Normalized Ratio serum values and potentiation of anticoagulant effects
-If concomitant therapy is necessary, the patient's INR should be more closely monitored

Concurrent use of chondroitin sulphate with anticoagulants might increase the risk of bleeding. Chondroitin may have interaction with warfarin activity.

Statement on Usage During Pregnancy and Lactation

Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mothers if the potential benefit to the mother justifies the potential risk to the fetus.

Administration during the first 3 months of pregnancy must be avoided.

Adverse Effects / Undesirable Effects(Glucosamine)
Cardiovascular:

Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.
Central nervous system: Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).
Gastrointestinal: Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.
Skin: Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

Overdose and Treatment

No cases of accidental or intentional overdose are known. The animal acute and chronic toxicological studies indicate that toxic effects and symptoms of toxicity are not likely to occur, even after high overdoses.

Storage Conditions

Store below 30°C in dry place.

Dosage Forms and Packaging Available

One box containing 5.7g x 14 sachets.

Shelf Life

3 years.

Name and Address of Manufacturer & Product Holder

AV Manufacturing Sdn.Bhd (667760-V)
Lot 10621 (PT16700), Jalan Permata 2, Arab Malaysian Industrial Park, 71800 Nilai, Negeri Sembilan, Malaysia

Date of Revision of Package Insert

Jun 2025 Ver. 01

Size PI: 150*130mm